

ARJUN VARMA

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EDUCATION

Columbia University

Master of Science in Data Science

Coursework: Probability and Statistics, Applied Machine Learning, Data Visualization, Analytics in Action

New York, NY

Aug 2025 - Dec 2026

Vellore Institute of Technology

Bachelor of Technology in Electronics and Communication Engineering, 4.0/4.0 (WES)

Coursework: Artificial Intelligence with Python, Data Structures and Algorithms, Web Mining

Recipient of the Special Achiever Award and a merit scholarship

Vellore, IN

Jul 2018 - May 2022

WORK EXPERIENCE

ZS Associates

Pune, IN

Advanced Data Science Associate Consultant

Feb 2025 - Jun 2025

- Worked in the Performance Analytics, Forecasting, and Data Science teams for a Fortune 500 healthcare client; collaborated with US-based stakeholders
- Built and deployed an organization-wide analytics + ML platform consolidating multiple data sources to surface real-time KPIs by territory and product; partnered with PMs and marketing heads for a >10B revenue oncology portfolio
- Drove adoption by 1,000+ sales reps and HQ leaders, replacing Excel reports and cutting prep time from days to minutes
- Piloted a retrieval-augmented LLM to turn FDA approval documents into concise briefs for the commercial teams

Decision Analytics Associate Consultant

Jul 2024 - Jan 2025

- Led a 5-member team on a strategic initiative to overhaul legacy business rules and modernize processes, saved ~50 hrs/mo and improved first-pass quality to >99%. Coordinated across other vendor teams to ensure uniform measurement of metrics
- Built and productionized Positive-Unlabeled (PU) learning models at a Fortune 500 organization to systematically infer missing categorical labels in transactional data, significantly expanding dataset coverage and market robustness for customer journey analysis; implemented automated model drift checks and unit testing to ensure long-term reliability
- Secured a top ~10% finish in a company-wide hackathon; selected for a lateral transfer into the Data Science vertical and completed a 4-month ramp
- Received *Client Comrade award* for client outcomes and feedback

Decision Analytics Associate

Feb 2022 - Jun 2024

- Engineered PySpark/SQL pipelines integrating multiple data sources to deliver brand performance across multiple products to senior leadership while working on the Performance Analytics Team of a blockbuster Oncology drug with revenue >\$4B
- Defined patient-cohort inclusion/exclusion rules robust to missing/miscoded fields; yielded consistent, audit-ready analytics
- Drove reporting and ad-hoc analytics that surfaced care gaps and market opportunities, informing key brand strategies across multiple new launches and changing market conditions
- Promoted to Associate Consultant in 4 cycles (typical: 5) via accelerated performance and received *Expert Associate and Insight Illuminator awards* for outstanding performance

PROJECT EXPERIENCE

Biliary Tract Cancer (BTC) early detection using ML – ZS Associates

Jan 2025 - May 2025

- Led development of an early detection model to predict monthly BTC diagnoses from a pool of 250M patients, addressed critical 45-day claims data delay; improved model performance by combining medical risk factors with advanced clustering techniques (K-means, Gaussian mixtures, NLP-based event clustering). Methods/results were presented at an industry conference

Agricultural Product Standardization & Risk Detection – Columbia University

Aug 2025 - Oct 2025

- Built an AI-assisted, retrieval-augmented generation (RAG) product-classification system for a Series-B East African agtech, processing 7M+ transactions for credit-risk assessment; hybrid rules + GPT-4 achieved **99%** holdout accuracy. Deployed a real-time REST API and dashboard for contract-compliance and environmental-risk alerts.

TECHNICAL SKILLS

Programming: Python, SQL, C++, R

Analytics & ML: Machine Learning, Deep Learning, Data Engineering, Extract Transform Load

Big Data & MLOps: PySpark/Spark, Databricks, MLflow, AWS (S3, EMR, Athena, SageMaker)

Libraries: pandas, NumPy, scikit-learn, matplotlib, BeautifulSoup, SHAP, PyTorch

Tools: Git, Jupyter/VS Code, Cursor IDE, Jira/Confluence